Chen-Fu Chien, Hung-Kai Wang, & Wen-Han Fu

Industry 3.5 Framework of an Advanced Intelligent Manufacturing System: Case Studies from Semiconductor Intelligent Manufacturing

(Received Jan 4, 2017; First Revision Apr 5, 2017; Second Revision Jun 9, 2017; Accepted Jun 29, 2017)

Abstract

This paper proposes that Industry 3.5 is a hybrid strategy that enables intelligent manufacturing with use of the big data analytics and digital decision-making processes of Industry 4.0 and the existing manufacturing system of Industry 3.0. The proposed strategic framework integrates leading technologies such as the Internet of Things, big data analytics, resource allocation, and optimization to develop an advanced intelligent manufacturing system framework. To determine the feasibility and validity of the proposed Industry 3.5 framework, this study reviewed a number of empirical studies concerning semiconductor manufacturing. The results indicated that the proposed Industry 3.5 is viable and can be used as an effective roadmap and reference to empower the migration of various industries to Industry 4.0 through adoption of advanced intelligent manufacturing, big data analytics, flexible decision-making, and optimization.

Keywords: Industry 4.0, Industry 3.5, Advanced Intelligent Manufacturing System, Big Data Analytics, Digital Decision-Making, Intelligent Fab

1. Introduction

With the rapid development of the Internet and intelligent technology, industrial applications integrating electronic networks, the cloud, and the use of intelligent devices are becoming increasingly prevalent. These developments compel industrial firms to grasp uncertainty easily and to respond rapidly. Globally, mobile and networked technology development has been accelerated by trends such as Design for X, social networking, the Maker subculture, three-dimensional (3D) printing, and the Internet of Things (IoT). For the manufacturing industry, these trends break the traditional boundaries of supply chains and manufacturing; thus, numerous emerging markets and innovative business models are springing up. Moreover, leading companies are all competing for dominance of industrial value chains in the future.

Many countries wordlwide have proposed manufacturing strategies: Germany proposed Industry 4.0 (Kagermann et al. 2013); the U.S. government promoted reindustrialization and established the Ad-vanced Manufacturing Partnership (AMP) plan to

Chen-Fu Chien is a Tsinghua Chair Professor in the De- partment of Industrial Engineering and Engineering Ma- nagement at National Tsing Hua University. (Corre- sponding Author) E-mail: cfchien@mx.nthu.edu.tw Hung-Kai Wang is an Assistant Professor in the Depar- tment of Industrial Engineering and Systems Mana- gement at Feng Chia University.
Wen-Han Fu is a Ph.D. Candidate in the Department of Industrial Engineering and Engineering Management at National Tsing Hua University.
DOI:10.6656/MR.201807_37(3).ENG105

move high-level manufacturing back to the United States (Executive Office of the President of the United States 2014); Japan proposed “Industry 4.1J”; South Korea promoted “Industrial Innovation 3.0”; and Mainland China implemented the “Made in China 2025” strategy to enhance Chinese compe- titiveness by upgrading China from a quantity manu- facturer to a quality manufacturer. Global compe- tition in manu- facturing has sparked the development of global advanced manufacturing technology, which is deconstructing and reconstructing the industrial value chain.

Germany’s Industry 4.0 involves three visions: The first entails collecting and analyzing big data re-lated to production processes through IoT; the second involves upgrading cyber-physical systems (CPSs) to intelligent manufacturing platforms that integrate all production blocks in the value chain; and the third entails developing advanced manufacturing systems to support flexible decisions in mass personalization. However, the deployment of Industry 4.0 cannot be realized overnight. Since the 1970s, the use of infor-mation technology and automated production systems has been regarded as a third industrial revolution; re-cently, this has been labeled Industry 3.0 (Kagermann et al. 2013). The development process of Taiwan’s manufacturing industry can be divided into three phases: A. the capacity development phase (1984- 2001); B. the information system integration phase (2002 to 2008); and C. the manufacturing intelligence phase (from 2009 to the present)(Chien et al. 2016). The first phase commenced in 1980, when the Hsinchu Science Park was established to drive the development of high-tech industries and related manufacturing. The second phase started from the

collapse of the Internet bubble in 2001, during which period Taiwan's high-tech industries faced the pressure of labor costs continuously increasing and wanted to integrate information systems into automated manufacturing systems; such integration can facilitate increasing productivity, improving production quality, saving labor costs, and increasing profitability. The third phase began after the impact of the 2008 global financial crisis; high-tech industries gradually entered an era of low profits accompanied by the maturation of different types of electronic products. Manufacturing companies sought opportunities to update from automation to intelligent production to enhance competitiveness.

For most countries worldwide, minimizing unemployment and revitalize the economy are general objectives, while advanced manufacturing is a strategic goal. The first two visions of Industry 4.0, big data and click-and-mortar, are means goals, whereas the third vision, flexible decisions and inte- lligent production, is the fundamental goal. Taiwan's most effective manufacturing strategy is to achieve means goals within its current manufacturing support environment. By using low-end disruptive innovation

facturing systems. It was hoped that this renaissance would reshape the US manufacturing industry, pull manufacturing companies back to the United States, drive the US economy, and improve the country's competitiveness (Executive Office of the President of the United States 2012).

AMP 2.0 was established in 2014 (Executive Office of the President of the United States 2014). At that time, the US national strategy had three main directions: A. advanced sensing, controls, and plat-forms for manufacturing; B. visualization, infor- matics, and digital manufacturing; and C. advanced materials manufacturing. The first direction mainly addressed control and optimization technologies, sen- sor and measurement technologies, and associated system architectures and platforms. The second direction explored digital manufacturing, digital representations, integrated information systems, and big data analysis. The third direction considered composite materials, biological manufacturing, and reprocessing of key materials. AMP 2.0 analyzed the current US manufacturing technology for these three directions, revealed some technological deficiencies, and issued corresponding recommendations for im-

in its existing manufacturing systems, Taiwan can achieve the means goal of big data analysis and system integration first. This can enable partial imple- mentation of flexible decision-making as envision in Industry 4.0. By these initial achievements, Taiwan can push the boundaries of a high-mix, low-volume production strategy to the uttermost limits, at which point mass personalization can be achieved. There- fore, the authors propose Industry 3.5 as a mixed strategy (Chien et al. 2016) between Industry 3.0 and Industry 4.0.

This study proposes an advanced intelligent manufacturing system (AIMS) framework, which integrates IoT, big data analytics, resource scheduling, and optimization technology to meet the research and development needs of Taiwan's semiconductor indus- try and addresses practical problems characteristic to the daily operations of the foundries; accordingly, the proposed framework can be used for formulating long- term research and development strategies and developing modular solutions. In addition, practical problems in semiconductor manufacturing were taken

provement.

as examples to test the validity of the framework. Whereas Industry 4.0 is still being developed, the Industry 3.5 framework proposed in this study has already been applied in real-world intelligent manufacturing. The proposed framework can be used in other industries to facilitate industrial progress.

2. Advanced Intelligent Manufacturing System Trend

2.1 US Manufacturing Renaissance

In 2012, the National Science and Technology Commission proposed an industrial and manu- facturing renaissance in the United States. This was to be initiated by the development of a national strategy for the future of advanced intelligent manu-

In March 2014, AT & T, Cisco, General Electric, IBM, and Intel established the Industrial Internet Consortium; their stated purpose was to break technology barriers and promote the integration of the physical and digital worlds. This consortium published the Industrial Internet Reference Architecture (IIRA)(Lin et al. 2015); the IIRA is an open architectural template and methodology with standard protocols; it applies to a wide range of industries, and it can drive industry interoperability. Architects who follow the IIRA can work with appropriate technologies and follow standardized development processes. The industrial internet archi- tecture should be considered from the perspectives of business, usage, function, implementation, security, connectivity, integration in technology, and data ma- nagement, analysis, and control.

2.2 Germany's Industry 4.0

Industry 4.0 was proposed by the German government in 2013 (Kagermann et al. 2013). Industry 4.0 differs from the three earlier stages of industrialization. The invention of the steam engine in the late eighteenth century led the first industrial revolution. The mass production of electric power in the early twentieth century led the second revolution. In the 1970s, programmable logic controllers led the third industrial revolution by enabling automation and access to production information. Germany's vision of Industry 4.0 emphasizes that the fourth industrial revolution will be based on CPS and IoT to enhance computerized, digitized, and intelligent manufacturing. As production methods become more efficient, the idea of a smart factory can become a reality.

Most existing manufacturing entails using batch production, but Industry 4.0 proposes CPS to

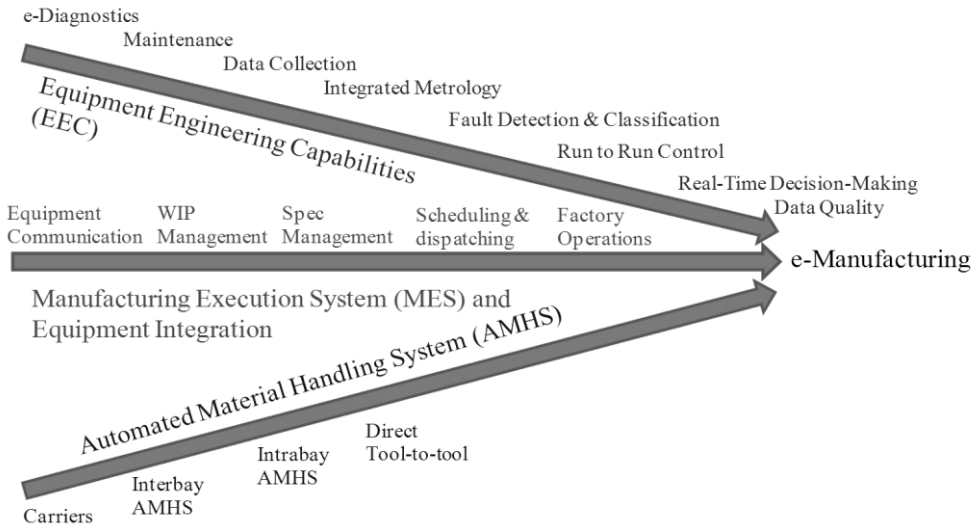

Figure1 Electronics Manufacturing Roadmap Source: SEMATECH, International (2003)

link the concept of individual items in a supply chain. All companies in that supply chain, from manufacturing to marketing and services, will be integrated to produce small volumes of individualized products that meet the exacting needs of particular customers.

Nevertheless, network information security must also be included in the standardized design of Industry 4.0; thus, constructing a secure reference architecture for the IoT might be necessary. However, delivering such a secure and standardized design necessitates addressing numerous challenges such as engineering technology, human quality and training, and cost-effectiveness.

For example, Siemens of Germany is one of the world's leading manufacturing systems companies. Siemens provides customized integrated systems and services for customers, resulting in the concept called Totally Integrated Automation (TIA) (Siemens 2015). Each TIA system integrates management, operation, and control systems from the enterprise level to the manufacturing level. TIA has been widely used in various types of enterprises including wood processing, metal cutting, food processing, and even elevator system development. One version of TIA is called SIMATIC IT(Siemens 2013), an integrated model involving a business level, a manufacturing execution system (MES), and a control level. The business level integrates enterprise resource planning, product lifecycle management (PLM), supply chain management, and other systems. The MES integrates product quality control and analysis systems that survey data at every point from raw materials entering the system to the output of the product. The control level is based on a real-time dispatching system, and integrates programmable logic controllers, distributed control systems, and open plat- forms for communication.

The Bosch Group of Germany is a technology and service provider. The group is divided into four business units, namely transportation solutions,

industrial technology, consumer goods, and intelli- gent technology for energy and construction. To participate in Industry 4.0, Bosch has established an Industry 4.0 concept-based production plant in Hom- burg. This production line integrates humans, pro- ducts, and machines into flexible and high-volume manufacturing production lines. The production line is profitable for highly customized products, even when the number of products in an order is only one. Bosch production line modules (Bosch 2015b) use production rules to produce useful information. Their major features are A. rapid data integration and implementation, B. multi-machine production data analysis, C. production goal-oriented information communication, D. rapid extension of customized rules to all machines, and E. continuously updated software solutions.

To anticipate future IoT trends, Bosch also developed an IoT-related module (Bosch IoT Suite)(Bosch 2015a), which contains the following five parts: A. device management (to link machines with central control systems); B. business rule mana- gement (to allow customers to set their own business rules that affect business decisions); C. business pro- cess management (to transfer business rules esta- blished in management system to other enterprises' own systems); D. big data analysis (to analyze the vast quantities of data generated from the IoT in order to derive useful rules to influence business decisions, with prediction functions to forecast future customer demand); and E. rights management (to manage the identities of all users and permissions in IoT). The IoT module connects the aforementioned five functions to customers, manu- facturers, and product designers. Customers can evaluate the product quality and send feedback to manufacturers; product designers can communicate with the manufacturer and revise products on this platform. This IoT module has been applied by Innova Electric, a vehicle manufacturer, and by firms in its supply chain.

2.3 Electronic Semiconductor Industry Road-map

In 2003, SEMATECH proposed three paths for semiconductor plants to move toward electronic foundry status (SEMATECH, International 2003): equipment engineering capabilities (EECs), MESs and equipment integration, and automated material handling systems (AMHSs), as shown in Figure 1. EECs include electronic diagnostic equipment operation, maintenance, data collection, integrated measurement, error detection, error classification, batch control, and real-time decision analysis; MES includes device-to-device communication, inventory management, inspection management, and scheduling and dispatching rule analysis; AMHS includes inner and outer tracks and direct machine-to-machine transmissions.

EECs include quality control and optimal con-trol parameters of machines. Quality control methods in product manufacturing systems include quality function deployment (QFD), failure mode and effects analysis (FMEA), inspection planning (IP), design of experiments (DOE), and statistical process control (SPC) (Inman et al. 2013). QFD focuses on converting customer requirements into the appro- priate quality requirements; FMEA emphasizes that defects must be detected before mass production; IP is the overall planning of the scheduled inspection activities; DOE is aimed at discovering and solving problems with few tests, short cycles, and low costs; and SPC is designed to focus on real-time monitoring and timely corrections. Each method has distinct advantages. An application based on the chara- cteristics of a different industry may greatly improve the product quality. In addition, machine learning is a type of algorithm for automatically analyzing data, obtaining the patterns of those data, and predicting unknown data. A computer running machine learning software can perform independent analysis rather than calculating according to a default algorithm. Machine learning theory in the manufacturing sector, especially in semiconductor manufacturing, has a wide range of applications including virtual metro- logy, fault detection, predictive maintenance, and run-to-run control (Castillo 2002).

MES requires a valid scheduling decision sup-port system that can limit the actual production issues to arrange shipments in the order of production prior-ity for different production machines. Semiconductor scheduling problems can be divided into six catego- ries: batching problems, problems with auxiliary resources, problems with multiple orders per job, scheduling of cluster tools, scheduling approaches for individual work areas, and job shop problems (Mönch et al. 2011). A production line must be scheduled by a decision support system to balance the load of the machine operations, to increase the capacity of operators and machine equipment utili-zation, and to reduce the waiting times of goods at different stages and the inventory costs of on-line products. Traditionally, mathematical programming

models and heuristic algorithms are used to optimize scheduling for different production lines.

AMHSs have been widely used in 12-inch wafer manufacturing plants and replaced human handling of materials. A typical AMHS for a conventional 8-inch wafer plant has crane tracks enabling automatic handling of wafer carriers to storage machines. AMHS tracks can be categorized as inter-bay and intra-bay. An inter-bay track spans multiple regions within a plant; it is analogous to a highway. An intra-bay track is contained within one region, servicing different machines; it is analogous to a local road between small towns. Various papers have discussed priorities for inter-bay and intra-bay vehicles transported by different dispatching rule decisions (Agrawal and Heragu 2006). However, IoT-based Industry 4.0 AIMSs will enable each batch of goods to determine its own path and communicate with every other batch in the future. Such pathfinding would be a great challenge for traditional AMHSs, which use central dispatching systems to determine shipment routes and priorities.

2.4 Smart Factory

Yoon, Shin, and Suh (2012) proposed a conceptual framework for a ubiquitous factory, which integrates a layer's life cycle, a ubiquitous product life cycle information highway, an application system, an agent layer, and a shop floor layer. This architecture connects product design, manufacturing, raw material and resource management, man–machine communication, and equipment–sensor rela- tionships. Lee (2015) proposed 5C architecture in smart factories, with five levels: A. smart connection (the ability to link sensors, self-awareness, and pre- diction); B. conversion (the ability to communicate between the components and the machine ditch system, which enables the machine to diagnose its own status); C. cyber (organization, communication, and maintenance between the fleets); D. cognition (interaction between decision-makers and decision support systems to achieve optimal decisions); and E. configuration (sending the knowledge of deci-sion-makers to the machine as feedback for subse-quent production decisions). Kang et al. (2016) speci-fied eight main areas for future smart factories: A. an actual situation integration system, B. cloud manu- facturing, C. big data analytics, D. networking, E. smart sensors, F. 3D printing, G. energy, and H. holo-grams.

Wang et al. (2016) listed six technical characteristics of a smart factory that differ from a traditional factory: A. diversified resources instead of reserved limited resources; B. dynamic production lines instead of a fixed route path; C. integrated net-work connections instead shop floor control; D. deep integration instead of end mill sites separated from the systems; E. organizational control instead of individually controlled independent machines; and F. information processing driven by big data analysis instead of isolated information processing. Thoben, Wiesner, and Wuest (2017) summarized Industry 4.0

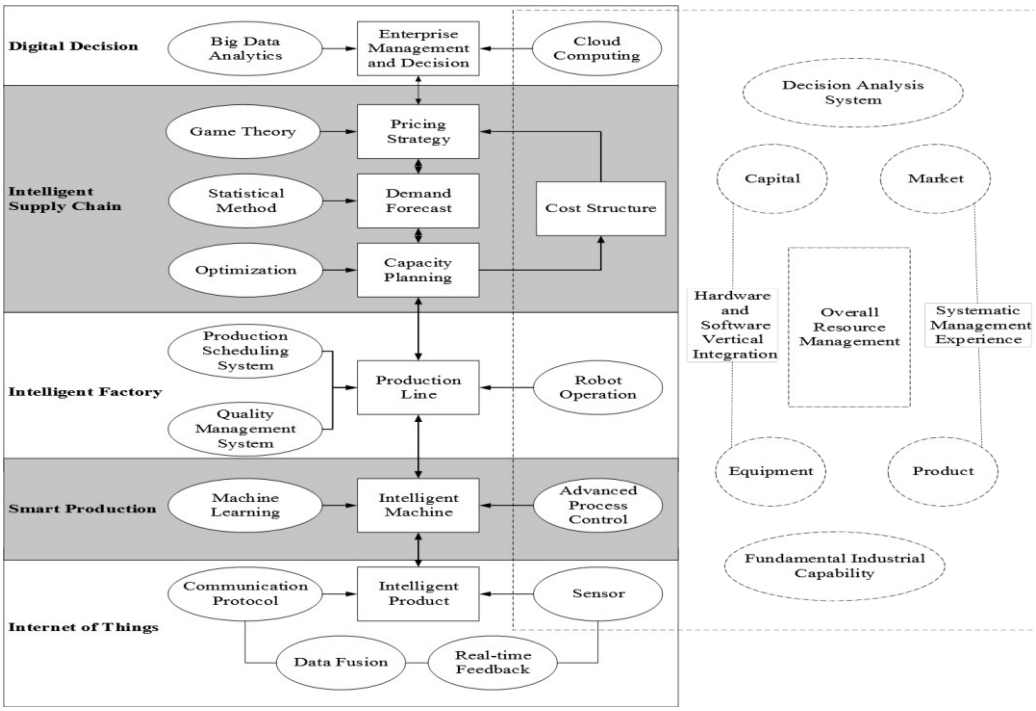

Figure 2 Advanced Intelligent Manufacturing System Framework

Source: Summarized by the Authors.

and manufacturing intelligence on three levels ranging from macroscopic to microscopic: A. cyber- physical logistics system, B. safe human–robot interaction, and C. video surveillance as a service. Related studies have reported on topics including technical standards (including interfaces), data analytics, data security, data quality, sensors and actuators, reference models, visualization, service marketplaces, app marketplaces, and requirements engineering. These can be considered a basis for the future development of smart factories.

The US Industrial Internet Consortium and Germany's Industry 4.0 initiative both plan for virtual and physical network connections to promote efficiency in production systems, but they differ in various details that reflect the different positions of these two countries in the global economy. The US software and Internet sectors are ahead of the rest of the world; therefore, the Industrial Internet Con- sortium emphasizes software services and hopes to use the construction of their industrial Internet to enhance the competitiveness of US manufacturing. Germany is a world leader in high-tech manufacturing; Germany's Industry 4.0 initiative hopes to develop intelligent factories directly to maintain the country's advantage. Other countries also study other aspects of intelligent manufacturing suitable for their respective industrial advantages. Therefore, in light of Taiwan's current industrial development, this study proposes an AIMS framework suitable for Taiwan in the third section. In the fourth section, we report on case studies of semiconductor manufacturing industry to illustrate the proposed framework, which can be a blueprint for the future development of intelligent manufacturing systems in Taiwan.

3. Advanced Intelligent Manufacturing System Framework

The progress of science and technology has subjected advanced manufacturing firms to global competition. Social networks have enhanced the product customization business model. Market de-mand fluctuates violently, and product life cycles are thus short. Hence, manufacturers must develop low-volume, high-mix customized production models. However, innovative AIMSs and management deci-sion models are necessary to design products and ser-vices that account for user experience and user deci-sion preferences. Production plans must consider production speed, flexible production, and the requirements of low-volume and high-mix products, economies of scale, production efficiency, green pro- duction, and sustainable development. Future manu- facturing firms need intelligent monitoring systems that integrate supply chains, enterprise product design, development, production, product life cycles, link procurement, outsourcing, distribution, and sales value chains. This integration should be accomp- lished through big data analysis and optimization tools. Plans should be adjusted flexibly in accordance with the latest information, to achieve automated production optimization and decision- making quality. Through these methods, the high-tech manufacturing industry can enhance its overall business performance and lead the development of new business oppor- tunities.

This study presents an Industry 3.5 strategy for an AIMS framework. This AIMS is explored on five network levels: digital decision, intelligent supply chain, intelligent factory, smart production, and IoT

(Figure 2). In a traditional manufacturing system, information can only be transferred along set paths. In this AIMS, each level can interact with any other level through the network by Internet of Things or cloud transmission technology. Thus, diverse paths of information exchange are available between elements within the various levels. A comprehensive resource management method is used to operate, manage, evaluate, and adjust the whole manufacturing process.

3.1 Digital Decision

The first step of enterprise development is to develop a general strategy and clarify short-term, medium-term, and long-term visions and plans. In the big data era, information can be collected easily, but direct and meaningful decision support is often difficult. Traditionally, decision-makers consider their own subjective experiences and make decisions that seem right to them. This method is no longer suitable; existing manufacturing systems can hardly have managers execute precise decisions quickly; digital decision-making has replaced human decision- making (Chien 2014). The supply chains of manu- facturing enterprises produce millions of zettabytes of structured and unstructured data (1 ZB = 10^{21} bytes). However, the ability to deal with these data is insufficient; thus, the effective utilization rate of the data is less than 5%. The information processing of manufacturing supply chains and corporate strategies must be transformed and upgraded to use industrial data collection, cloud computing, and other methods to deal with zettabytes of information in order to obtain effective knowledge, which can then be used to develop strategic guidelines.

Big data technologies can transform raw data into practical, useful information. Big data technolo-gies can construct integrated platforms for predictive analysis and visualization, which are key capabilities for intelligent manufacturing firms that hope to de-velop sustainable innovation in the Industry 4.0 era (Lee, Kao, and Yang 2014). This large-scale data analysis cannot be completed with a personal com-puter or server. Large-scale cloud computing servers can access the data and analyze them; cloud compu-ting is the best method for enterprises to obtain effec-tive information and to develop operational strategies.

Cloud computing, integrated with IoT, is now becoming one of the critical force multipliers of manufacturing. It may change traditional manufa- cturing business models, promote product innovation, enhance business strategy, and create an effective and synergistic intelligent factory network (Xu 2012). High speed and low cost are the most crucial chara-cteristics of cloud computing. In the development stage, the cloud can be a test environment to save money and time. In the acquisition stage, the cloud can help enterprises successfully integrate acquisi- tions to ensure a smooth transition. In the transfer phase, the cloud can facilitate resource allocation for effective deployment. If any emergency occurs, cloud computing can help decision-makers to respond

quickly and effectively, which can show corporate resilience and competitiveness.

3.2 Intelligent Supply Chain

Enterprises can obtain most valuable orders through big data analysis, and apply pricing strategies and price relationship analysis to determine the opti-mal prices of goods. For example, short-term demand in the electricity industry changes dramatically. Uni-form prices would increase the average costs of electricity, but time-of-use strategies based on game theory methods (Yang, Tang, and Nehorai 2013), which obtain Nash equilibria by backward induction, can be used as a demand-side management approach to influence user needs. This can improve customer satisfaction and reduce costs for the company.

The future demand for goods can be predicted by statistical methods after the product price has been determined. For the semiconductor industry and for other capital-intensive industries with high capacity utilization, interchangeable products and similar foundry services can be provided by other companies, and this is because various companies have access to the necessary technology; thus, sales competition is fierce. Multiple-generation SMPRT diffusion models (Chien, Chen, and Peng 2010) can facilitate demand forecasting. A model that considers the seasonal fac-tors, market growth rates, prices, repeat purchases, and technology substitutions can improve demand forecasting accuracy to provide a company with more accurate capacity planning.

A firm's industrial engineering department must decide on a production plan, and distribute production capacity to each foundry for production according to the predicted demand. Capacity planning is usually performed under demand uncertainty. A minimax regret strategy (Chien and Zheng 2012) can increase the productivity of semiconductor manufacturing capacity utilization and capital efficiency. When each possible outcome of a multistage demand forecast is taken into account, a robust capacity expansion plan can be created to minimize losses of production capacity due to overcapacity and shortages.

Production planning and distribution processes are related to a company's production costs. The cost of production in one quarter will affect the company's profitability and pricing strategy in the next quarter. Therefore, the company must continuously optimize pricing strategies, demand forecasting, capacity plan-ning, and cost structures (Chien, Chen, and Peng 2010) to help itself make profits in a market full of uncertainty.

3.3 Intelligent Factory

TSMC classified the development of smart factories into three stages: anthropomorphic, un-manned, and ultra-human. In early smart factories, computers had learned management rules designed for factories with human workers. In the next stage of smart factories, the automated systems replace human

workers. Finally, the development of an AIMS that integrates the intelligence of human workers allows an ultra-human smart factory to outperform a factory staffed by average human workers (Chien 2017). Specifically, after a merchandise order is assigned to an ultra-human factory, the automated production lines and robots operate efficiently and deliver the product rapidly, stably, and at a low manpower cost. The paramount mission of the robots is to improve efficiency and reduce tact time, an essential measure of production line efficiency that is defined as the ratio of total effective production time to the amount of production output.

A study reported on final state control with a high-speed positioning control method; this was used for limiting industrial robot acceleration and velocity, and incorporated a linear matrix inequality for effec-tive time management (Yabuki et al. 2014). The choice of a robot can be quantified by objective attri- butes (including cost, reliability, load capacity, and repeatability of positioning accuracy) and subjective qualitative attributes (including vendor service con-tracts, training, human–computer interface, and programming flexibility) through the analytic hierar-chy process and other analysis methods (Rao, Patel, and Parnichkun 2011). In recent years, robot design has been progressing in functions, and the cost has been declining. Even managers who had previously

low-volume products to deliver large numbers of personalized products. Production management that relies on the human brain and personal experience has its limits. An AIMS offers flexible decision rules for smart production. These rules are condensed from manufacturing intelligence extracted from domain experts. Each machine in the AIMSs can determine any necessary decision immediately, considering yield, delivery, and productivity; thus, each machine can adjust its internal production program parameters in accordance with product quality to maximize profitability. For example, an advanced process con-trol model can adjust production parameters in accordance with previous batch product shipments (Castillo 2002).

Advanced process control uses measurement data from a batch of products that has just been pro-duced, to feedback optimal parameters to machines in the production line. Thus, each machine can be ad-justed dynamically in accordance with measurements of recently finished products. Each machine is fre-quently tuned with optimal parameters, and the factory maintains process quality and stability for some target variables. Compared with conventional SPC, in which an abnormal signal appears only when a quality problem is detected, advanced process control is more effective because it executes real-time monitoring and adjusts machine production parame-ters immediately.

opposed robots are gradually starting to use robots to perform tasks in different manufacturing systems.

Factory memory can be described as a portfolio containing a variety of machine functions and pro-ducts. A production scheduling system is required to automatically assign products to different machines for production; such a system is effective if it can minimize machine idle time, product waiting time, and production cycle time. A typical scheduling deci-sion support system in a semiconductor factory is designed to minimize cycle time; the system also considers the limited waiting time of a shipment between work stations. Semiconductor wafers that are excessively delayed exhibit chemical changes, which may render the product useless (Chien and Chen 2007).

Production lines must always be controlled by the quality control system. If the output of the machine has abnormal quality, the product is a waste of raw materials, and is likely to delay the delivery of a satisfactory product to the customer. Although production and quality are two crucial goals, they are in conflict with each other. A study explored how quality control and production control have combined with each other in recent years (Yugma et al. 2015). For example, either run-to-run control or fault dete-ction and classification can be set as the most crucial constraint of a scheduling decision support system that simultaneously considers the production and quality problems. To perfect such AIMSs with this functionality will be a challenge in the future.

3.4 Smart Production

Modern production models are growing in complexity. AIMSs may produce high-mix and

Optimization of manufacturing process para- meters can adjust machines to maximize production efficiency. Conventional parameter optimization ap-proaches include genetic algorithms, simulated an- **nealing, particle swarm optimization, ant BM quantum computers leverage furniture** optimization, and artificial bee colony methods (Yusup, Zain, and Hashim 2012).

If the machines are equipped with sensors, the system can immediately collect operating status parameters, analyze the situation, and predict mach- ine health. If the diagnosis of the machine shows abnormalities, the system can automatically issue a warning to allow engineers to perform timely mainte-nance, to reduce the losses caused by abnormal behavior (including crashes). Smart production can monitor the status of the process and the diagnosis of the machine itself, to maintain high-quality produ- ction and improve efficiency.

3.5 Internet of Things

When each product on the production line is intelligent, the outputs of various sensors can be fused with information from different sources; the resulting data set can be analyzed. The final indi- vidual product can make its own suitable decisions at different points (Chien 2015). For example, a product will need to answer such questions as “When should I start processing?”, “What are the most suitable parameters for processing?”, or even “Where should I go after I complete processing, and why?” Whereas a traditional factory has centralized control of the production line, an AIMS endows the individual components with enough intelligence to decide their

processing times and their next destinations. However, numerous technique gaps must be overcome, such as the standardization of communication protocols, which will determine communication approaches

For example, the Overall Equipment Effective-ness method (Nakajima 1988) is used to calculate and evaluate the operational effectiveness of a production facility. Different evaluation indices can be developed between for different categories of resources within

Factories require internal coordination for diverse equipment components in the same system. Multi-agent service manufacturing collaborative architectures facilitate internal coordination of enterprise control features. These architectures facilitate management and control of distributed, automated, intelligent manufacturing networks. For connections between different companies, the current trend is to use a computer as a transit point. Different sensors collect data for later analysis. In the AIMS, the IoT makes the tools themselves capable of intelligent analysis. Components analyze and share information through a communication protocol to ensure the effective use of information. Ultimately, this saves resources and improves efficiency.

Among the current communication protocols for IoT, IBM MQTT and IETF CoAP are mainstream and competitive choices; they are likely to be the future standard protocols for communication on the IoT. A comparative study (Thangavel et al. 2014) showed that for a low packet loss ratio (the ratio of communication data loss during transmission), the delay of MQTT is lower than that of CoAP. By contrast, for a high packet loss ratio, the delay of MQTT is greater than that of CoAP. When the data size is small and loss rate is less than or equal to 25%, the additional traffic generated by CoAP to promote the reliability of the information is less than that of MQTT.

Currently, the different protocols cannot be unified; therefore, the IoT cannot build bridges between different systems. Because the morphology and quality of information from different sources are inconsistent, data must be processed through data fusion before analysis. Complex data processing is a major problem for practical applications. In the future, when unified standard communication protocols are developed, cross-system objects will be connected, data security and privacy will be improved, and the IoT will become a critical part of intelligent manufacturing systems.

3.6 Overall Resource Management

Considering the five Advanced Intelligent Manufacturing System levels shown in Figure 2, we can establish systematic resource management indica-tors. A manager can combine these indicators with

manufacturing plants, according to various industry backgrounds and production properties. Some evalua-tion indices for measuring intelligent manufacturing systems are as follows: Overall Wafer Effectiveness (Chien, Hsu, and Chang 2013) for identifying different types of losses, devices, photolithography techniques, exposure modes, and production fluctuations; Overall Usage Effectiveness (Chien, Diaz, and Lan 2014) for measuring overall efficiency and material losses caused by various factors, including tool type, formulation, material, process steps, and processing time; Overall Space Effectiveness (Chien, Hu, and Hu 2016) for the diagnosis of plant layout, efficient use of space, and overall space productivity; and Overall Resource Effectiveness (Chien, Chu, and Zhao 2015)for measuring the efficiency and effectiveness of resource usage.

4. Case Study

Currently, at the dawn of the Industry 4.0 era, the United States, Germany, and other advanced countries have strong industrial bases, high-tech manufacturing advantages, and software capabilities. Furthermore, Mainland China and Southeast Asia have strong low-end advantages in cheap labor and mass production. Taiwan currently occupies an intermediate position. The core competitiveness of Taiwan is its rich domain experience accumulated in manufacturing management and production processes. Taiwan should use Industry 3.5, a disruptive conceptual innovation, to advance in the global supply chain as the intelligent manufacturing era progresses.

In this transformative period of upgrades from Industry 3.0 to Industry 4.0, Taiwan should strengthen its basic industrial technology and vertically integrated analysis capability continuously. In addition, Taiwan should systematize and digitize its management knowledge through big data analysis, networking, and other technology, to develop its ability to produce solutions that meet the customized needs of consumers. Accordingly, Taiwan can occupy a niche market position with many advantages unavailable to Industry 3.0. Taiwan will have these advantages before other countries' manufacturing systems upgrade to cope with the future challenges of

personal management experience to develop Industry 4.0. analytical techniques and intelligence systems for internal and external resources owned by the enterprise; these can be used to conduct an overall inventory analysis. The four steps of the Plan – Do – Check – Act (PDCA) resource management cycle can enhance utilization efficiency and productivity. PDCA can optimize resource allocations and

practical problems in Taiwanese semiconductor manufacturing. The case study presents an AIMS perspective on reliability and validity, big data yield analysis, demand forecasting, intelligent scheduling, advanced process control, wafer bin map defect diagnosis, and material handling systems. The AIMS can develop modular solutions to enhance the

synergize capital performance- resources competitiveness (Chien and Hu 2011; Chien and Lin 2016). chain (Figure 3).

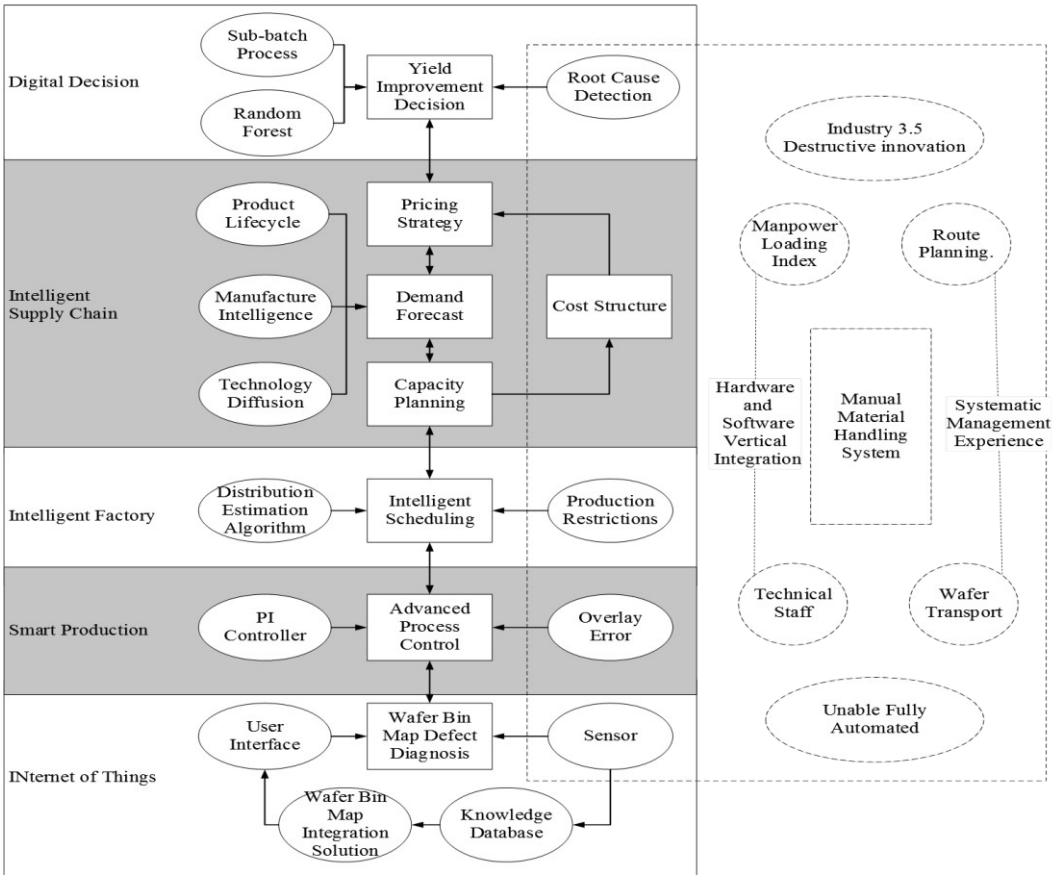

Figure 3 Semiconductor Advanced Intelligent Manufacturing System Framework
Source: Summarized by the Authors.

4.1 Digital Decision for Big Data Analytics in Yield Enhancement

The semiconductor manufacturing process is one of the most complex manufacturing systems in existence. It includes several hundred process steps, such as physical vapor deposition and chemical vapor deposition, photolithography, etching, and ion implantation. The wafer, the most commonly used semiconductor material, is the basis of all processes; integrated circuits are fabricated on wafers. This fabrication produces zettabytes of data. As integrated circuit product cycles shorten and prices decrease, root cause detection becomes increasingly critical to yield enhancement. A semiconductor manufacturer with superior yield has a competitive advantage. The trend toward completely automated wafer processing, with increasingly large amounts of relevant infor- mation and increasingly complex structures, exa- cerbates the difficulties of fault detection for the entire production system.

Batch wafer processing typically leads to low variance in product quality characteristics in the same batch and high variability between batches. Batch processing can be accelerated with advanced pro- cesses on the production line. (Chien and Chuang 2014) presented a sub-batch processing model, com- bined with detection of the root causes of wafer defects by random forest fixing collinearity.

The present research involved empirical obser- vation of one of Taiwan's leading wafer fabrication foundries for root cause detection through sub-batch process systems. The dependent variable was wafer probing test yield; the possible factors were recorded parameters in the process and the test wafer acceptance. The test data contained 18 batches of 447 wafers with 3,919 factors. Among these factors, 938 parameters were related to the machine tool, 469 were related to the process recipe, 469 were related to the module, 469 were related to the supply, and 1,574 were related to wafer acceptance test. The first four factors were nominal, whereas the fifth factor was a numerical factor. Because some raw data were missing, 1,641 items were retained after a preliminary data preparation process, of which 729 were ma-chine- related, 203 were recipe-related, 102 were module-related, 41 were supply-related, and 471 were test-related. After the key factor screening, model construction, and evaluation, two significant factors were excavated. First, factor No. 260 of the reaction chamber formulation in Phase No. 667 was con- sistently better than factor No. 259, indicating that the reaction chamber for No. 259 was a root cause of yield loss. Second, in wafer acceptance test No. 99, circuit probe yield was reduced. This thus proves the effectiveness of the proposed detection architecture.

Data mining and big data analytics can be used to improve wafer yield, increase productivity, fore-

cast demand (Chien and Hsu 2014), and discover the potential effects of factors from vast quantities of information. Based on these effects, analysists can establish data-driven predictive models to control these factors to further improve the yield, enabling foundries to develop manufacturing strategies. Based on the results of big data analytics, factories can establish domain databases. When new processes or new products are proposed, managers can use historical data and models as references for finding problems and key sites quickly.

4.2 Intelligent Supply Chain for Manufacturing Intelligence in Product Demand Forecasting

The semiconductor industry is a capital-intensive industry. Effective productivity affects a semiconductor manufacturing company's capital expenditures and profits. However, because semiconductor manufacturing involves make-to-order production, long-term prediction models based on historical data tend to be affected by technological changes and demand uncertainty; thus, historically based prediction becomes quite difficult (Chien, Chen, and Peng 2010; Chien, Dou, and Fu 2018).

In practice, most companies still use traditional models of product demand prediction. Such models depend on regional sales decisions based on consumer reactions, personal domain knowledge, and market experience. The sales of products produced from such traditional models are susceptible to the bullwhip effect. The bullwhip effect refers to the tendency of a supply chain to amplify demand variability, because of the restriction of demand forecasts; the production line must therefore hold safety stock. Thus, the inventory analysis involves the cumulative demand of downstream industries; fore- casting the demand of upstream industries is also restricted, because manufacturers gradually progress from downstream to upstream levels.

Manufacturing intelligence is to extract useful information, inferred from the products and supply chain data, and to support manufacturing strategy decision-making. (Chien, Chen, and Peng 2010) pro-posed an SMPRT model, based on product life cycle and technology diffusion theory, to construct a semiconductor product manufacturing intelligence architecture for predicting demand. To evaluate the effectiveness of the method, a Taiwanese semi- conductor company was studied. A total of 36 quarters of data were used to obtain the parameters of the proposed model, and nonlinear least-squares methods were used to test the accuracy of predictions.

Seasonal factors are first considered in predic-tion models, because the demand for semiconductor products such as computers and televisions is affected by different seasons. The second factor is the market growth rate, which is used to describe the structure of the market and economic environment; this rate affects customer behavior. The third factor is the price; the price of a new generation of products tends to affect customers' willingness to purchase the

products. Customer behavior tends to change with expectations of future prices. For semiconductors, the initial product launch prices fall rapidly, but then in later stages, prices equilibrate to a steady state. The fourth factor is repeated purchasing; if the customer does not purchase sufficient quantities in one transaction, that customer will make purchases later. Finally, technology substitution causes products with new technology to diffuse into an existing market and replace some products with older technology. Diffusion and replacement must be considered as long as products made with old technology compete with newer products.

In the empirical study, the proposed method predicted the characteristics of three new tech- nologies B, C, and D with high accuracy. The absolute percentage error (APE) values for the technologies were 4.6%, 4.6%, and 0.1%. However, for technology A, an older generation technology, the APE value was 97.1%. Some products with tech- nology A are still in production to supply spare parts for outdated equipment. Demand may be increased by other technologies. However, the production of products with technology A is already low; therefore, the foregoing demand factors are relatively weak.

In the manufacturing supply chain, demand forecasting is an essential task. If the predicted de-mand is lower than the actual demand, productivity cannot be fully utilized, causing the machine to be idle and thus wasting resources. If the actual demand is lower than the predicted demand, production is too low to meet the demand, resulting in the loss of orders and reputation. The use of manufacturing intelligence is to obtain an accurate forecast of demand for products, which leads to pricing strategies that induce customers to buy the products. Such strategies also facilitate capacity planning. Thus, an advantage in terms of cost structure can be obtained. However, the effects of demand factors may change with different products or market conditions. If such changes occur, the existing model can be modified or reconstructed based on market conditions, to maintain the com- pany's mastery of product demand.

4.3 Intelligent Factory for Semiconductor Intelli- gent Scheduling

Managers in the semiconductor manufacturing industry are expected to deliver products that satisfy Moore's Law (Moore 1965); that is, the number of transistors per unit area of electricity is expected to double every 12–24 months. The technology require-ments also change according to process technology advancements and market shrinkage. With increas-ingly sophisticated process technology, all varieties of processes are pushing closer to the physical limits, and the targets of process stability control are becoming more rigorous. Both yield requirements and process control stability requirements are in- creasing dramatically in semiconductor manufa- cturing.

Due to increasing process complexity and shrinking transistor sizes, more constraints must be considered in the semiconductor scheduling problem, such as a maximum waiting time limit, machine states, different processing times of the same work-piece on different machines, set times of each work, and arrival times of each workpiece. Among all considerations, the maximum waiting time limit is paramount. If a wafer waits for processing for any length of time exceeding the maximum waiting time, its surface may be contaminated, and chemical reac-tions may change the electrical conductivity of the wafer; such a wafer cannot be processed further and must be scrapped, which incurs an exorbitant loss for the company.

However, the environment of a semiconductor production line changes rapidly. Scheduling must be solved completely within a few minutes, to meet the real-time needs of the production line. Traditional mathematical programming models cannot be solved completely within a reasonable time if the scale of the problem expands. (Wang, Chien, and Gen 2015) used a distribution estimation algorithm to assist a semiconductor foundry to obtain satisfactory solu-tions within minutes; this thus increased machine capacity utilization, improved productivity, and ultimately enhanced that foundry's profitability.

Distribution estimation algorithms are heuristic algorithms, which differ from conventional algo-rithms such as genetic algorithms. A probability distribution estimation algorithm can apply a probability matrix to produce virtual progeny to search for new solutions. Traditional heuristic search algorithms are relatively blind; a probability matrix can be learned and updated according to the high-performing solutions in each generation, which enables more statistical learning (Larrañaga and Lozano 2001).

The proposed method of this case was practi-cally applied to a large-scale actual semiconductor plant scheduling problem. This problem involved 300-500 work pieces arranged on 50-70 sets of ma-chine stations; 10 different possible scenarios were analyzed. Various disruptions were considered, including process contamination, new generations of products, urgent work orders, importation of new machines to a newly built factory, and natural disas-ters (such as earthquakes and water disasters) causing machine downtime. The distribution estimation algo-rithms could outperform traditional genetic algo-rithms by 47%, and they could reduce delay in 69% of cases. Finally, the distribution estimation algo-rithms could be combined in a semiconductor produc-tion scheduling decision support system to optimize treatment planning. These algorithms can automati-cally and instantaneously adjust production lines in accordance with current conditions.

The aforementioned scheduling decision sup-port system is traditionally implemented as a central-ized plant-wide control mechanism. A traditional plant responds to an organic condition or process

problems by using its central scheduling system to rearrange operations for optimal scheduling results. In the future, Industry 4.0 will dominate manufactur-ing; thus intelligent factories, intelligent machines, and intelligent products may be equipped with intelli-gence. Each product can automatically determine its new destination as soon as it completes its processing on any machine. Intelligent products will make individual decisions to communicate with all ma-chines to confirm the status of machines, and deter-mine the most suitable machine for processing. How-ever, the optimization of individual decisions cannot be guaranteed to optimize scheduling for the entire foundry. Therefore, communication and coordination between the timing of individual decisions and scheduling decision support systems will be a vital issue for Industry 4.0 intelligent scheduling systems.

4.4 Smart Production for Semiconductor Manu- facturing Advanced Process Control

The factors in the semiconductor manufactur-ing process are complicated (e.g., temperature, pres-sure, wafer flatness, process accuracy, input parame-ter settings, and measurement errors). Various random interference values affect the process, making the process difficult to stabilize. Process control methods can ensure the process is stabilized at a target value.

Traditional SPC uses control charts to monitor whether a serious variation has occurred in the pro-cess. The system is not adjusted if it stays within predetermined boundaries. If the process exceeds the boundaries, then machines must be stopped and checked until the outliers have been found. However, this approach cannot be applied to semiconductor manufacturing; semiconductor production must be continuous and no downtime can be tolerated. If the upper and lower control limits are set too narrowly, false alarms will increase and the time of wafer out-put will be prolonged, delaying delivery and causing substantial losses for the semiconductor foundry. Thus, basic SPC cannot solve generated drift in semi-conductor processes.

Advanced process control can use differences in premeasured batches of wafers. The system can consider uncontrollable interference values to deter-mine process parameters. The process can achieve its required target stability value. In a conventional control system, control parameters must be set, and the settings of those parameters deeply affect the outcomes. Such control methods typically make as-sumptions regarding the model inputs and outputs (IO model) and the controller, use a certain type of interference distribution, set the same combination of control parameter values, and investigate long-term stability to find a steady-state region.

Semiconductor industries undergo rapid prod-uct changes; a semiconductor manufacturing system has on average 300-500 manufacturing system pro-cesses. The process factors are uncountable; stability before film processing affects the yield after each layer of the product is fabricated. Interference disturbance is difficult to estimate and predict for

each batch, even with mathematical models. A control system can assume a fixed interference value for a long control parameter in a steady-state situation, but that assumption does not match the actual process conditions. Process interference values often vary with time. The question of how to determine the parameters of the controller is crucial. Likewise, the process of dynamically adjusting the control parame-ters from batch to batch is crucial.

In semiconductor manufacturing, photolitho- graphy processes are among the most expensive and most critical processes. The stability of the process depends on the stability of the line width. However, **as technology improevs and product features shrink,**

Batch control considers controllable and un- controllable manufacturing processes together. Con- trollable processes may be estimated in advance using a realistic IO model. The calculated input value can be substituted into the process to calculate the corresponding output value. For uncontrollable pro-cesses (interference values), a process IO model can-not assess the relevant factors. The total IO value of such a model is the actual process output value. After the difference value calculation process outputs, the target value, that value is substituted into the control-ler for calculations. The controller determines the number of input values. The difference value for a desired process can be reduced. The calculated value substituted into the process is called feedback control or compensation.

the line width diminishes and the tolerable degree of variation is also reduced; hence, the process stability control grows ever more difficult. In addition to controlling the output value of the line width of the outside layer and the rear layer processed in the previous step, the system must control layer overlay errors (i.e., errors regarding the relative alignment of layers). These errors may affect the functionality of the semiconductor elements, such as electrical conductivity and insulation. To minimize errors, the semiconductor elements can be produced to achieve the desired functionality.

machines are expensive, the lithography step is a bottleneck in the production flow. Using a batch method can control the value of an output control process to some target value. By reducing the process variation, one can increase the stability of the process. Heavy machines substantially reduce the rework ratio of defective products, and this can reduce the bottle-neck effect of lithography, thus reducing the cycle time of the wafers.

Overlay error can mostly be divided into two parts: intra-field and inter-field. Intra-field overlay errors are caused by misalignment between the reticle and the wafer. The overlay requires a mask, a light source, and a filter lens. The center point of the refer-ence measurement for each exposure field or shot center must be controlled to minimize error. In-ter-field overlay errors are caused by misalignment of the field between the wafer and the exposure field, or by a wafer being subjected to certain deformation temperature and pressure errors. Although measuring machines can measure a total coverage error, this error cannot identify the impact of each individual factor. An overlay error may be composed of a verti-cal and a horizontal overlay error (Chien, Chang, and Chen 2003).

A previous case (Chien et al. 2014) focused on rapidly changing disturbance values in a semiconduc-tor manufacturing process. Such disturbances are difficult to estimate. In some cases, the dynamic adjustment of proportional-integral control parame-ters may be used. The estimated disturbance values of current batches may be used to adjust PI control parameters to determine the optimum number of inputs.

An empirical study was conducted at a Taiwan-ese dynamic random access memory (DRAM) manu- facturing facility. The proposed method reduced average overlay errors in the X direction by 13% (with a root mean squared error of 8%) and in the Y direction by 23% (with a root mean squared error of 16%). This result can be extended to semiconductor processing in other regions.

However, feedback control does not have enough information. An immediate problem is that machine conditions or process conditions can change at any time. Furthermore, previous installments are not immediately measured to provide feedback infor-mation. Currently, shipment problems have emerged. Thus, Industry 4.0 must build predictive big data models of virtual measurement, using the predicted values for feedforward control. Each machine must have sufficient intelligence to adjust its process parameters conveniently to ensure that the process can be stabilized at a target value. Finally, in the aforementioned advanced manufacturing systems, quality control is integrated into the real-time motion control process and is thus a crucial component of the decision support systems. Therefore, automation and intelli- gence will improve production lines.

4.5 Internet of things Application for Wafer Bin Maps Defect Diagnosis

Wafer bin maps hold crucial information regarding abnormalities in the semiconductor manu- facturing process. A wafer map can be used to model the causes of low yield. Typically, a semiconductor plant has a predetermined pattern for its wafer maps, and engineers mostly judge those maps visually. However, due to wafer miniaturization and the increased volume of integrated circuits, some wafer maps may have inconsistent graphics for tooling styles, wafer rotation, and mold density failure modes. Humans find it difficult to make reliable and valid judgments of such maps.

To improve the yield of semiconductor manu- facturing processes, Liu and Chien (2013) integrated spatial statistical tests, cellular neural networks, adaptive resonance theory, neural networks, and mo

ment invariant methods to construct wafer bin maps
for an intelligent defect diagnosis system. The system

does not have the AMHS described previously due to aging equipment, it cannot implement fully auto-mated transport.

consists of three modules: a graphical user interface, wafer map aggregation solution, and knowledge data-base. This system involves control and data flows. The control flow originates from the client; it can be used to control wafer bin map system parameters and obtain a wafer pattern type from the customer inter-face. The data flow is opposite to the control flow in terms of direction and path. Three interface modules provide information through the graphical user inter-face. Engineers can extract useful information for support of manufacturing strategy decisions to achieve manu-facturing intelligence. To produce a wafer bin map, the database information is gathered into a wafer pattern system; the results are summa-rized in a report to the user and recorded in the knowledge base.

Based on expertise and historical defect data, the described study by Liu and Chien (2013) selected four types of performance indicators (purity, diversity, specificity, and efficiency) to model and assess 12 types of wafer bin map patterns. A control group comprised a ward method, nonhierarchical k-means clustering method, self-organizing map, spectral clus-ter method, and clustering method proposed by Chien et al. (2002). The empirical analysis indicated no significant difference in the diversity; comparing the pattern noise and sensitivity revealed that the pro-posed method was superior to the conventional meth-ods in purity. In some specific cases, the proposed method was worse than the ward and k-means clustering methods. Purity and diversity are the main topics of pattern detection in practice. The proposed method could effectively diagnose the defects shown on wafer bin maps and import real-time libraries for subsequent machine learning and knowledge manage-ment.

In the study by Liu and Chien (2013), the sys-tem sensed graphical data, determined the precise ex-tent of each product, and then traced the causes of abnormalities in the manufacturing process. After the classification had been analyzed and summarized, intuitive reports were delivered through the graphical user interface and recorded in the knowledge base for future reference. By using human-computer inter-acttion, this semi-automated model can effectively improve the decision-making speed at the foundry end, and can gradually accumulate domain know- ledge of defect diagnosis. When mature IoT hardware and software are available, the system can be upgraded to Industry 4.0 standards, and it can enable the machines in intelligent factories to diagnose themselves.

4.6 Industry 3.5 Strategy for Manual Material Handling System

Intelligent production is a mixed strategy be-tween the current Industry 3.0 and the future Industry 4.0. However, many factories cannot implement intelligent production in practice for technical or cost reasons. For example, if a 200-mm fabrication plant

To meet the needs of certain real-world facto-ries, Chien, Chou, and Yu (2016) proposed a novel manual material handling system, which delivers functionality analogous to automated material han-dling functions. Based on an existing 200-mm fab, a manual material handling system was developed on the basis of a disruptive innovation. Technical staff were taught to manually simulate a virtual material handling system to provide transportation for an artificial wafer lot and to work systematically, thereby increasing productivity.

Mathematical optimization models for route planning typically have one of two goals: to minimize the total distance traveled in all directions, or to mini-mize all manpower needs for all routes. A route plan-ning method was proposed to minimize the technical staff travel distance and transportation time. In addi-tion, a manpower loading index was introduced to evaluate the quantities of technical staff required in the proposed MMHS. To estimate the validity of the proposed MMHS, a simulation environment was developed based on empirical data; the environment involved scenarios to compare different transporta-tion needs. The results showed that the proposed method had practical feasibility. The company has actually adopted and verified the proposed method.

The company in that case was not able to apply automated production to the state-of-the-art standards of Industry 3.0 because of its aging factory facilities. Nevertheless, it was able to analyze and optimize data by increasing plant capacity for the management of manpower allocations; this adaptation can be re-garded as an example of an Industry 3.5 manufactur-ing strategy. This provides a reference for hu-man-based traditional industries to improve produc-tion systems. However, when these traditional firms expand a new plant's production lines or construct a production sys- tem in the future, they can take the proposed advanced manufacturing system framework as a blueprint to upgrade gradually to an Industry 4.0 intelligent factory.

5. Conclusion

Taiwan's manufacturing industry is familiar with the high-level division of labour, the pursuit of returns to scale to reduce the cost of production and manufacturing, and the technology of Industry 3.0, but Industry 4.0 vision, strategic objectives, value proposition, and core competency requirements are different. This study presents the Industry 3.5 frame-work, particularly the notion of AIMSs, as a technol-ogy roadmap. Different firms operating at different stages of semiconductor manufacturing were taken as real-world cases to illustrate intelligent decision sup-port systems involved in AIMS.

Future research directions include the develop-ment of machine learning and artificial intelligence to

make automated factory equipment learn continuously from big data, enabling instant decisions for self-regulation of intelligent manufacturing systems. Each individual intelligent factory machine has the capability for smart decisions and the capability to communicate with other production items and processes instantly when monitoring the surrounding environment. Machines must be able to find problems in situ at any time, and either exclude those problems or notify managers through instant automatic alerts. Machines must know how to deal with notifications from process engineers. The system must optimize individual local decision-making. Big data analytics and optimization techniques must be developed to make intelligent production more flexible.

With the United States and Germany and other advanced countries returning to manufacturing and developing automated intelligent manufacturing platforms, and with the traditional mass production of Industry 3.0 shifting to emerging countries with abundant cheap labor, Taiwan should develop Industry 3.5 as a hybrid strategy integrating Industry 3.0 and Industry 4.0. Taiwan should elaborate a blueprint for an "Industry 3.0 – Industry 3.5 – Industry 4.0" strategy. In addition, Taiwan should apply big data analytics in manufacturing and AIMSs for disruptive innovation within the existing Industry 3.0 economy, and it should integrate enterprise resources and advantages to establish dedicated upgrade strategies and intell- igent manufacturing technology strategies in order to seize a share of the industrial upgrade market. More- over, Taiwan should systematize and digitize the core abilities of flexible supply chain scheduling and yield enhancement in order to integrate PLM and thus achieve considerable customization flexibility; digitizing those types of knowledge can enable meeting consumer demand rapidly and intelligently. Additionally, Taiwan can continuously improve its fundamental industrial technology, enhance its analysis capabilities with software and hardware equipment, and improve its vertical integration; doing so can enable Taiwan to widen the gap with emerging countries and maintain its relative advantages.

References

Agrawal, Gaurav K. and Sunderesh S. Heragu (2006), "A Survey of Automated Material Handling Systems in 300-mm Semiconductor Fabs," *IEEE Transactions on Semiconductor Manufacturing*, 19(1), 112-120.

Bosch (2015a), "Harnessing the Power of the Internet of Things - The IoT for the Extented Enterprise,"research report,Bosch Software Innovations.

---- (2015b), "Production Rules Configurator - Rules Based Analysis of Production Data with Connected Manufacturing Solutions,"research report,Bosch Software Innovations.

Castillo, Enrique Del (2002), *Statistical Process Adjustment for Quality Control*. New York: John Wiley & Sons.

Using Advanced Process Control with Dynamically Adjusted Proportional-Integral R2R Controller," *IEEE Transactions on Automation Science and Engineering*, 11(2), 473-484.

Chien, Chen-Fu, Ting-Hao Lin, Qiao-Wen Liu, Cheng-Yung Peng, Shao-Chung Hsu, and Chia-Chi Huang (2002), "Developing a Data Mining Method for Wafer Binmap Clustering and an Empirical Study in a Semiconductor Manufacturing Fab," *Journal of the Chinese Institute of Industrial Engineers*, 19(2), 23-38.

----, Kuo-Hao Chang, and Chih-Ping Chen (2003), "Design of a Sampling Strategy for Measuring and Compensating for Overlay Errors in Semiconductor Manufacturing," *International*

---- and Shih-Chung Chuang (2014), "A Framework for Root Cause Detection of Sub-Batch Processing System for Semiconductor Manu- facturing Big Data Analytics," *IEEE Tran- sactions on Semiconductor Manufacturing*, 27(4), 475-488.

----, Alejandra Campero Diaz, and Yu-Bin Lan (2014), "A Data Mining Approach for Analyzing Semiconductor MES and FDC Data to Enhance Overall Usage Effectiveness (OUE)," *International Journal of Computational Intelli- gence Systems*, 7(sup2), 52-65.

---- (2014), *Decision Analysis and Management: A UNISON Framework for Total Decision Quality Enhancement*, 2nd ed. Taipei: Yeh-Yeh Book Gallery.

---- and Chia-Yu Hsu (2014), *Data Mining and Big Data Analytics*. Taipei: Future Career Publishing Corporation.

---- (2015), "The Crisis and Turns of Internet of Thing,"Interview by Yu-I Lee, *Harvard Busi- ness Journal* (June), 150-

---- and Chien-Hung Chen (2007), "A Novel Timetabling Algorithm for a Furnace Process for Semiconductor Fabrication with Constrained Waiting and Frequency-Based Setups," *OR Spectrum*, 29(3), 391-419.

----, Yun-Ju Chen, and Jin-Tang Peng (2010), "Manufacturing Intelligence for Semiconductor Demand Forecast Based Diffusion and Product Life Cycle," *International Journal of Production Economics*, 128(2), 496-509.

Journal of Production Research, 41(11), 2547-2561.

----, Pei-Chun Chu, and Lizhong Zhao (2015), "Overall Resource Effectiveness (ORE) Indices for Total Resource Management and Case Studies," *International Journal of Industrial Engineering*, 22(5), 618-630.

----, Che-Wei Chou, and Hui-Chun Yu (2016), "A Novel Route Selection and Resource Allocation Approach to Improve the Efficiency of Manual Material Handling System in 200-mm Wafer Fabs for Industry 3.5," *IEEE Transactions on Automation Science and Engineering*, 13(4), 1567-1580.

---- and Chih-Han Hu (2011), "Total Resource Management Framework: An Empirical Study of Semiconductor Fab," *Journal of Quality*, 18(6), 581-607.

---- and Jia-Nian Zheng (2012), "Mini-Max Regret Strategy for Robust Decisions in Semiconductor Manufacturing," *Journal of Intelligent Manufacturing*, 23(6), 2151-2159.

----, Chia-Yu Hsu, and Kuo-Hao Chang (2013), "Overall Wafer Effectiveness (OWE): A Novel Industry Standard for Semiconductor Ecosystem as a Whole," *Computers & Industrial Engi- neering*, 65(1), 117-127.

----, Ying-Jen Chen, Chia-Yu Hsu, and Hung-Kai

----, Chih-Han Hu, and Yi-Fen Hu (2016), "Overall Space Effectiveness (OSE) for Enhancing Fab Space Productivity," *IEEE Transactions on Semiconductor Manufacturing*, 29(3), 239-247.

---- and Kuo-Yi Lin (2016), "Constructing the Framework of Semiconductor Overall Fab Performance Indices for Total Resource Management," *Journal of Management & Systems*, 23(4), 451-474.

---- , Kuo-Yi Lin, Jiuh-Biing Sheu, and Cheng-Hung Wu (2016), "Retrospect and Prospect on Operations and Management Journals in Taiwan: From Industry 3.0 to Industry 3.5," *Journal of Management and Business Research*, 33(1),
Wang (2014), "Overlay Error Compensation 87-103.

---- (2017), Mönch, Lars, John W Fowler, Stéphane "Strategic Capacity Planning
'Industrial 3.5' ," *Harvard Business Review Complex Chinese Edition*, 128(April), 46-54.

----, Runliang Dou, and Wenhan Fu (2018), Production: Decision Modeling under Demand Uncertainty," *Applied Soft Computing*, 68(July),

Dauzère-Pérès, Scott J Mason, and Oliver Rose (2011), “A Survey of Problems, Solution Techniques, and Future Challenges in Sche- duling Semiconductor Manufacturing Opera- tions,” *Journal of Schedu- ling*, 14(6), 583-599.

Moore, Gordon (1965), “Cramming More Compo- 900-909.

Executive Office of the President of the United States (2012), “A National Strategic Plan for Advanced Manufacturing,”research report,The President’s Council of Advisors on Science and Technology of the United States.

nents onto Integrated Circuits,” *Electronics*, 38(8), 114-117.

---- (2014), “Accelerating U.S. Advanced Manu- facturing, AMP2.0 Steering Committee Report,”research report,The President’s Council of Advisors on Science and Technology of the

Nakajima, Seiichi (1988), *Introduction to TPM: Total Productive Maintenance*. Cambridge, MA, US: Productivity Press.

Rao, R. Venkata, Bhisma K. Patel, and Manukid Parnichkun (2011), "Industrial Robot Selection Using a Novel Decision Making Method Consi- dering Objective and Subjective Preferen- ces," *Robotics and Autonomous Systems*, 59(6),
United States. 367-375.

Inman, Robert R., Dennis E. Blumenfeld, Ningjian Huang, Jingshan Li, and Jing Li (2013), "Survey of Recent Advances on the Interface between Production System Design and Quality," *IIE Transactions*, 45(6), 557-574.

SEMATECH, International (2003), "E-Diagnostics and EEC Guidance,"research report,SEMA- TECH, Austin, Texas.

Siemens (2013), "SIMATIC IT Intelligence Suite v2.1: Functional Overview,"research report, Siemens AG.

Kagermann, Henning, Johannes Helbig, Ariane Hellinger, and Wolfgang Wahlster (2013), "Recommendations for Implementing the Strategic Initiative INDUSTRIE 4.0: Securing the Future of German Manufacturing Industry,"research report,Forschungsunion.

Kang, Hyoung Seok, Ju Yeon Lee, SangSu Choi, Hyun Kim, Jun Hee Park, Ji Yeon Son, Bo Hyun Kim, and Sang Do Noh (2016), "Smart Manufacturing: Past Research, Present findings, and Future Directions," *International Journal of Precision Engineering and Manufacturing-Green Technology*, 3(1), 111-128.

Larrañaga, Pedro and Jose A Lozano (2001), *Estimation of Distribution Algorithms: A New Tool for Evolutionary Computation*. New York: Springer Science & Business Media.

Lee, Jay, Hung-An Kao, and Shanhu Yang (2014), "Service Innovation and Smart Analytics for Industry 4.0 and Big Data Environment," *Procedia CIRP*, 16 (June), 3-8.

---- (2015), "Smart Factory Systems," *Informatik-Spektrum*, 38(3), 230-235.

Lin, Shi-Wan, Bradford Miller, Jacques Durand, Rajive Joshi, Paul Didier, Amine Chigani, Reinier Torenbeek, David Duggal, Robert Martin, and Graham Bleakley (2015), "Industrial Internet Reference Architecture,"research report, Indu- strial Internet Consortium.

Liu, Chiao-Wen and Chen-Fu Chien (2013), "An Intelligent System for Wafer Bin Map Defect Diagnosis: An Empirical Study for Semicon- ductor Manufacturing," *Engineering Applica- tions of Artificial Intelligence*, 26(5), 1479-1486.

---- (2015), “Profit from Engineering Efficiency: Totally Integrated Automation Portal in Pra- ctice,”research report,Siemens TIA Portal.

Thangavel, Dinesh, Xiaoping Ma, Alvin Valera, Hwee-Xian Tan, and Colin Keng-Yan Tan (2014), “Performance Evaluation of MQTT and CoAP via a Common Middleware,”paper presented at the 2014 IEEE Ninth International Conference on Intelligent Sensors, Sensor Net- works and Infor- mation Processing (ISSNIP), Singapore (April 21-24).

Thoben, Klaus-Dieter, Stefan Wiesner, and Thorsten Wuest (2017), “‘Industrie 4.0’ and Smart Manufacturing–A Review of Research Issues and Application Examples,” *International Jour- nal of Automation Technology*, 11(1), 4-19.

Wang, Hung-Kai, Chen-Fu Chien, and Mitsuo Gen (2015), “An Algorithm of Multi-Subpopulation Parameters with Hybrid Estimation of Distri- bution for Semiconductor Scheduling with Constrained Waiting Time,” *IEEE Transactions on Semiconductor Manufacturing*, 28(3), 353-366.

Wang, Shiyong, Jiafu Wan, Di Li, and Chunhua Zhang (2016), “Implementing Smart Factory of Industrie 4.0: An Outlook,” *International Journal of Distributed Sensor Networks*, 12(1), 1-10.

Xu, Xun (2012), “From Cloud Computing to Cloud Manufacturing,” *Robotics and Computer-Integrated Manufacturing*, 28(1), 75-86.

Yabuki, Akinori, Hien Nguyen, Takashi Yoshioka, Kiyoshi Ohishi, Toshimasa Miyazaki, and Yuki

Yokokura (2014), "Final State Control Based on High-Speed Positioning Control Considering Acceleration and Speed Limits for Industrial Robot," paper presented at the 2014 10th France-Japan/8th Europe-Asia Congress on Mecatronics (MECATRONICS), Tokyo Metropolitan University (November 27-30).

Yang, Peng, Gongguo Tang, and Arye Nehorai (2013), "A Game-Theoretic Approach for Optimal Time-of-Use Electricity Pricing," *IEEE Transactions on Power Systems*, 28(2), 884-892.

Yoon, Joo-Sung, Seung-Jun Shin, and Suk-Hwan Suh (2012), "A Conceptual Framework for the Ubiquitous Factory," *International Journal of Production Research*, 50(8), 2174-2189.

Yugma, Claude, Jakey Blue, Stéphane Dauzère-Pérès, and Ali Obeid (2015), "Integration of Scheduling and Advanced Process Control in Semiconductor Manufacturing: Review and Outlook," *Journal of Scheduling*, 18(2), 195-205.

Yusup, Norfadzlan, Azlan Mohd Zain, and Siti Zaiton Mohd Hashim (2012), "Evolutionary Techniques in Optimizing Machining Parameters: Review and Recent Applications (2007-2011)," *Expert Systems with Applications*, 39(10), 9909-9927.